Project: **REAL ESTATE VALUATION (HOUSE PRICE PREDICTION)**

Problem Understanding:

The objective of this project is to analyze and prepare the Boston Housing dataset and develop a supervised machine learning model to predict house prices. The study explores how housing related features influence residential property values and evaluates the predictive performance of a Linear Regression model.

Dataset Description:

The Boston Housing dataset contains information about housing prices in different areas of Boston. Each row represents a residential area with environmental factors, and socio-economic characteristics.

- Number of observations/rows: **506**
- Number of features: **13**
- Target variable: **MEDV (Median house price in thousands of dollars)**

Example Features:

- **CRIM:** Crime rate per capita
- **RM:** Average number of rooms per dwelling
- **NOX:** Nitric oxide concentration
- **LSTAT:** Percentage of lower-status population
- **PTRATIO:** Pupil–teacher ratio

All features in the dataset are numerical, which makes the dataset suitable for regression analysis.

Data Preparation and Cleaning:

Dataset Exploration:

The dataset was loaded using scikit-learn and converted into a Pandas DataFrame for analysis. The structure of the dataset was examined by checking:

- First five rows of the dataset
- Number of rows and columns
- Data types of all features
- Summary statistics using the describe() function

The dataset consists of 506 rows and 14 columns (**13 features and 1 target variable**). Statistical analysis showed reasonable ranges, means, and distributions for all features.

Datatypes Found:

```
2]: CRIM        float64
    ZN          float64
    INDUS       float64
    CHAS       category
    NOX         float64
    RM          float64
    AGE         float64
    DIS         float64
    RAD        category
    TAX         float64
    PTRATIO     float64
    B           float64
    LSTAT       float64
    MEDV        float64
    dtype: object
```

The Describe() function: showing the mean, minimum, maximum, and standard deviation for each feature

[13]:

	CRIM	ZN	INDUS	NOX	RM	AGE	DIS	TAX	PTRATIO	B	LSTAT	MEDV
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	3.613524	11.363636	11.136779	0.554695	6.284634	68.574901	3.795043	408.237154	18.455534	356.674032	12.653063	22.532806
std	8.601545	23.322453	6.860353	0.115878	0.702617	28.148861	2.105710	168.537116	2.164946	91.294864	7.141062	9.197104
min	0.006320	0.000000	0.460000	0.385000	3.561000	2.900000	1.129600	187.000000	12.600000	0.320000	1.730000	5.000000
25%	0.082045	0.000000	5.190000	0.449000	5.885500	45.025000	2.100175	279.000000	17.400000	375.377500	6.950000	17.025000
50%	0.256510	0.000000	9.690000	0.538000	6.208500	77.500000	3.207450	330.000000	19.050000	391.440000	11.360000	21.200000
75%	3.677083	12.500000	18.100000	0.624000	6.623500	94.075000	5.188425	666.000000	20.200000	396.225000	16.955000	25.000000
max	88.976200	100.000000	27.740000	0.871000	8.780000	100.000000	12.126500	711.000000	22.000000	396.900000	37.970000	50.000000

Missing Value Analysis and Handling:

Identification of Missing Values

The dataset was examined for missing values across all features. No missing values were found in any column.

Missing values may occur due to:

- Incomplete data collection
- Errors during data recording
- System or sensor failures
- Human input mistakes

Risks of Ignoring Missing Data:

Ignoring missing values can negatively affect machine learning models. It may result in biased datasets, incorrect model training, and unreliable predictions. Improper handling of missing data can also increase the risk of underfitting or overfitting.

Handling Strategies

Although the dataset contained no missing values, standard handling techniques were demonstrated for completeness:

1. **Mean/Median Imputation:** Missing values are replaced with the mean or median of the feature to preserve dataset size.
2. **Row Removal:** Rows containing missing values are removed when the number of missing entries is very small and statistically insignificant.

Justification (What I did):

Here the mean and median imputation were work better as they preserve dataset size and reduce bias when missing values are minimal. Row removal was considered but not applied, as it can lead to information loss.

```
Before Handling Missing Values:
CRIM       0
ZN         0
INDUS      0
CHAS       0
NOX        0
RM         6
AGE        0
DIS        0
RAD        0
TAX        6
PTRATIO    0
B          0
LSTAT      0
MEDV       0
dtype: int64
```

After Handling missing values:

```
After Mean/Median Imputation:
CRIM       0
ZN         0
INDUS      0
CHAS       0
NOX        0
RM         0
AGE        0
DIS        0
RAD        0
TAX        0
PTRATIO    0
B          0
LSTAT      0
MEDV       0
dtype: int64
```

Outlier Detection and Treatment:

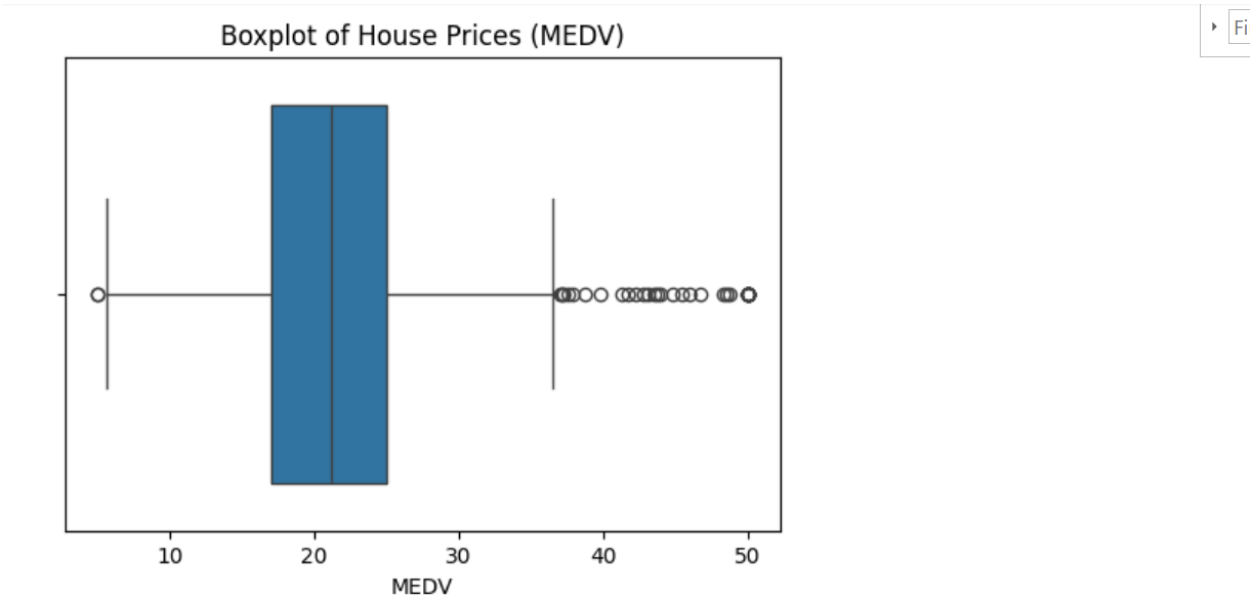
Outlier Detection:

Outliers were analyzed using two methods:

1. **Interquartile Range (IQR) Method:**
   This method identified high-value outliers in the target variable (MEDV).
2. **Z-score Method:**
   No outliers were detected using the Z-score method, as it is less sensitive to skewed data distributions.

Visual analysis using boxplots and distribution plots confirmed the presence of extreme high house prices.

Detected Outliers:



Impact of Outliers on Regression Models:

Outliers can impact the regression models as they bind it toward extreme values. This can reduce prediction accuracy, and lead to unstable model performance.

Outlier Treatment:

To reduce the influence of extreme values while preserving the dataset size, "**Capping**" was applied to the target variable. In this method, values above the upper IQR boundary were capped at the upper limit.

Other techniques such as log transformation and outlier removal were also considered; however, capping was selected as the final approach because it balances the data and retention.

Outliers Adjusted on key Features:

CRIM (Crime Rate):

This feature showed a highly right-skewed distribution with extreme values. Capping was applied to limit the effect of unusually high crime rates.

LSTAT (Lower Status Population %):

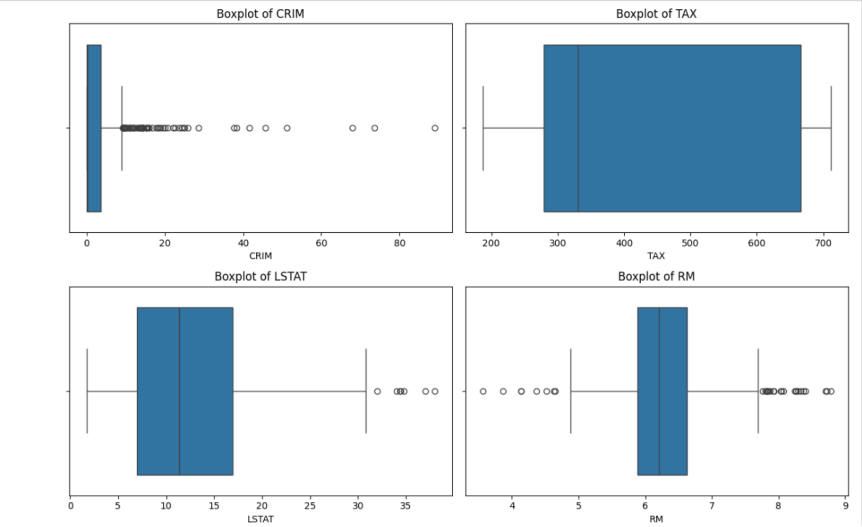
Extreme high values were present and capped to prevent disproportionate influence on house price predictions.

TAX (Property Tax Rate):

Although some high values were observed, capping was applied cautiously to reduce skewness while retaining meaningful variation.

RM (Average Number of Rooms):

This feature showed a relatively stable distribution. Since extreme values were meaningful and limited, no outlier treatment was applied



Final Cleaned Dataset:

After completing missing value analysis and outlier treatment, the dataset was finalized for modeling:

- No missing values remain
- Outlier influence has been reduced
- Dataset size remains unchanged
- Data is clean and ready for regression modeling

The cleaned dataset confirms that it is well structured and suitable for the next phase of model development.

[30]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222.0	18.7	396.90	5.33	36.2

Model Development and Evaluation

Train–Test Split:

The dataset was split into:

- **80% training data**
- **20% testing data**

Linear Regression Model:

A **Linear Regression** model was trained using the prepared dataset. The model learns a linear relationship between the input features and the target variable (MEDV).

- **Intercept:** Represents the baseline house price when all features are zero
- **Coefficients:** Each coefficient represents the expected change in house price for a one-unit increase in the corresponding feature, keeping all other features constant

Positive coefficients indicate features that increase house prices, while negative coefficients indicate features that decrease house prices.

[40]:

	Feature	Coefficient
0	CRIM	-0.503420
1	ZN	0.031927
2	INDUS	-0.003087
3	CHAS	1.322545
4	NOX	-12.395920
5	RM	2.916456
6	AGE	-0.012389
7	DIS	-1.178767
8	RAD	0.313512
9	TAX	-0.010158
10	PTRATIO	-0.750952
11	B	0.009848
12	LSTAT	-0.458090

Predictions:

43]:

	Actual	Predicted
0	23.6	27.737821
1	32.4	32.973482
2	13.6	14.555869
3	22.8	24.937748
4	16.1	18.169453
5	20.0	22.502476
6	17.8	18.087907
7	14.0	14.365679
8	19.6	22.012544
9	16.8	20.805081

Model Evaluation:

The model was evaluated using standard regression metrics:

- **Mean Squared Error (MSE):** Measures average squared prediction error
- **Root Mean Squared Error (RMSE):** Measures average prediction error in house price units
- **R^2 Score:** Measures the proportion of variance explained by the model

Results:

- **R^2 = 0.75**: The model explains approximately **75%** of the variation in house prices
- **RMSE = 3.5**: Predictions are reasonably accurate

The similarity between training and testing performance indicates that the model is **neither underfitting nor overfitting**.

48]:

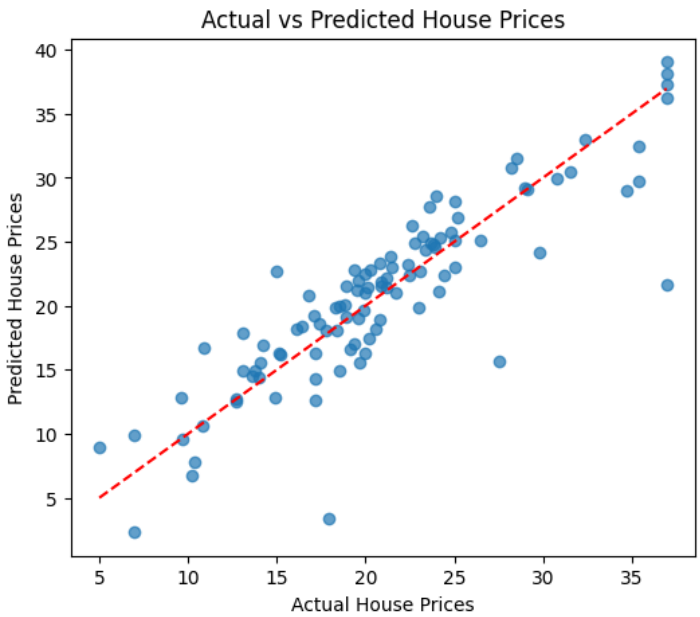
	Metric	Value
0	MSE	12.135836
1	RMSE	3.483653
2	R^2	0.752049

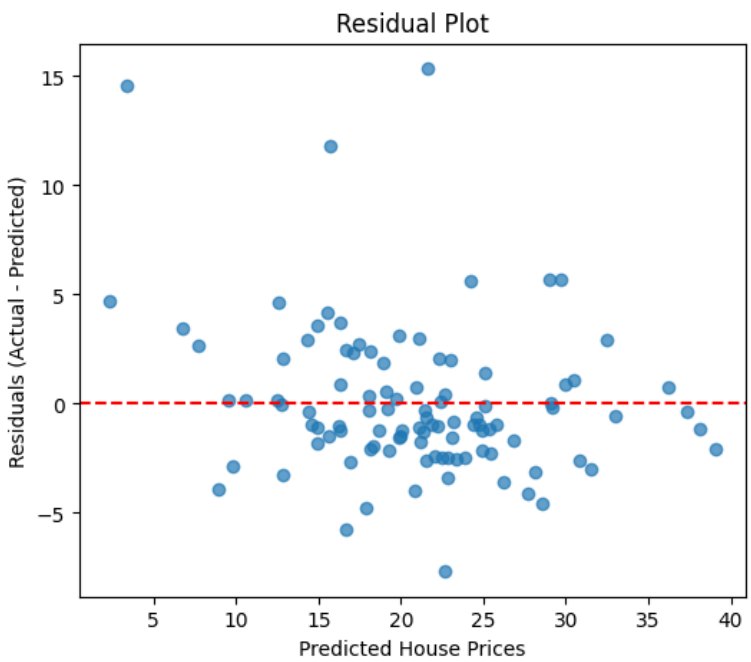
Visualization and Optimization:

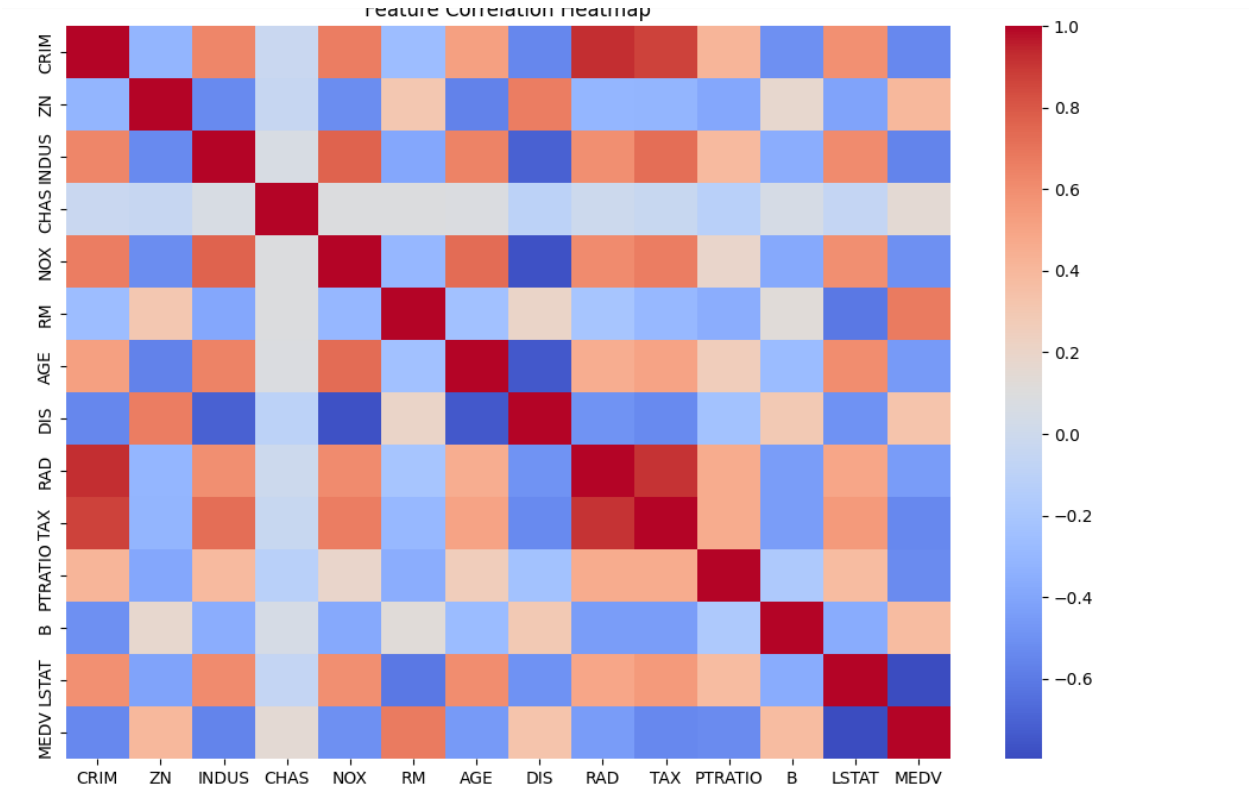
Visualization:

The following visualizations were used to evaluate model performance:

- **Actual vs Predicted Scatter Plot:** Shows predictions closely aligned with actual values
- **Residual Plot:** Residuals are randomly scattered around zero, indicating no systematic errors
- **Feature Correlation Heatmap:** Highlights strong relationships between features and the target variable


Actual vs Predicted House Prices
Predicted House Prices
Actual House Prices
40
35
30
25
20
15
10
5
5
10
15
20
25
30
35


Residual Plot
Residuals (Actual - Predicted)
Predicted House Prices
15
10
5
0
−5
5
10
15
20
25
30
35
40



Feature Scaling:

- Feature scaling was applied using **StandardScaler** to standardize all features to a common scale.
- After scaling, the model showed similar RMSE and R^2 values
- The result shows that the scaling did not negatively impact performance

```
Model Performance After Feature Scaling:
RMSE (Scaled): 3.484
R² Score (Scaled): 0.752
```

Results and Interpretation

The Linear Regression model successfully captures the relationship between housing features and house prices. Features such as RM and LSTAT play a significant role in prediction. Evaluation metrics and visualizations confirm strong predictive performance and good generalization.

Conclusion

In this project, the Boston Housing dataset was successfully explored, cleaned, and modelled using Linear Regression. The complete machine learning workflow was implemented, including data preprocessing, model training, evaluation, visualization, and feature scaling. The final model achieved strong predictive performance and provides a reliable foundation for real estate price estimation